

Log Scale Surprise Teacher Notes

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Medium	ID	log-scale-surprise

Big idea: Mathematical representations can make the same relationship look explosive, steady, or manageable.

Overview

Students compare linear and logarithmic graphs and see how scale choice changes visual interpretation.

Student Prompt

Inspect Log Scale Surprise Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Log Scale Surprise Stimulus A	Student-facing stimulus 1: Dataset: 1, 2, 4, 8, 16, 32, 64 over seven days. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Log Scale Surprise Stimulus B	Student-facing stimulus 2: Question: does the graph show runaway growth, steady multiplicative growth, or both?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Log Scale Surprise Reveal / Twist	Student-facing stimulus 3: Notation card: $\log_{10}(1000) = 3$ shows how a multiplicative change becomes an equal step on a log scale. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Mathematical representations can make the same relationship look explosive, steady, or manageable.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Log Scale Surprise Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Log Scale Surprise Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	How does mathematical representation shape understanding?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how mathematical representations can make the same relationship look explosive, steady, or manageable.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Use simple doubling data so students can see both the numerical and visual relationship.
- Let students describe each graph before teaching the term logarithmic scale.

Ready-to-Use Examples

- Dataset: 1, 2, 4, 8, 16, 32, 64 over seven days.
- Question: does the graph show runaway growth, steady multiplicative growth, or both?
- Notation card: $\log_{10}(1000) = 3$ shows how a multiplicative change becomes an equal step on a log scale.

Suggested Timing

Stage	Teacher move	Watch for
1	Give students a simple growing dataset.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Groups plot or inspect the same data on a linear graph and a logarithmic graph.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Students describe the story each graph seems to tell.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Groups identify which graph is better for different purposes.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief how mathematical representation affects knowledge claims.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Both graphs can be accurate, but they serve different questions.
- What does this scale make visually obvious, and what does it hide?

Common Misconceptions

- Students may treat a persuasive pattern as equivalent to proof.
- Students may trust intuition when the formal model uses a different structure.

Assessment Look-Fors

- Distinguishes intuition, empirical pattern, model, and proof.
- Explains why formal reasoning can conflict with first impressions.

Differentiation and Adaptations

- Support: use pre-drawn graphs.
- Challenge: ask students to decide which scale is more responsible for a given audience.

Extension Tasks

- Students find a real public graph and ask what scale would change the visual story.
- Compare scale choice with headline framing and dashboard design.

Related Activities

- Graph Caption Framing
- Base Rate Neglect Clinic
- Proof versus Pattern